



Air Quality Situation Report

Edmonton Symphony Orchestra Outdoor Concert · Hawrelak Park (Heritage Amphitheatre)

Thu, Sep 3, 2026 · 4:37 AM MDT

Air Quality Health Index (AQHI) — Edmonton

2.0

Current AQHI at Edmonton Lendrum (3.51 km from venue)
Forecast Thu 6:00 AM at Edmonton Lendrum (3.51 km from venue): **1.9**

Health Risk	AQHI	At-Risk Population	General Population
Low Risk	2.0	Enjoy your usual outdoor activities.	Ideal air quality for outdoor activities.

No active Environment Canada weather alerts for the venue.

Environmental Intelligence Summary

- At Edmonton Symphony Orchestra Outdoor Concert, temperature is 15.6°C and feels near 15.5°C. Winds are 8 km/h from NE, gusting near 19 km/h.
- The nearest current AQHI is 2 at Edmonton Lendrum, 3.5 km from the venue.
- The official Government of Alberta AQHI for Edmonton is 2 (forecast tonight: 3, tomorrow: 3).
- A gridded AQHI estimate blending official and community sensors (confidence: High, nearest contributing point 4.0 km away) puts the area near 1.
- The nearest air monitoring station (Edmonton Lendrum, 3.5 km) reports fine particulate matter (PM2.5) at 4.5 µg/m³.
- A nearby community sensor ('ACA_LaurierHeightsCL', 1.8 km) reads 5.6 µg/m³ PM2.5.
- Back-trajectory modeling (HRDPS wind fields, 6 h lookback) indicates the air currently at the venue traveled from the vicinity of a point 118.2 km to the SE, i.e. arriving from the SE.
- The nearest active fire detection (NASA FIRMS – VIIRS, last 2026-09-02) is 10.2 km E of the venue. The modeled back-trajectory passes within 3.2 km of this fire, making smoke transport to the venue plausible based on wind-field modeling.
- The AQHI forecast for Thu 6:00 AM is 2, based on Edmonton Lendrum, 3.5 km from the venue.
- Thunderstorm conditions appear in the hourly forecast beginning around Thu 12:00 PM.
- Overall operational environmental risk is HIGH. Primary concerns are precipitation, thunderstorm.

Recommendations

- Confirm lightning monitoring, shelter, pause and evacuation procedures.
- Prepare drainage, electrical protection and wet-weather controls.
- Environment Canada AQHI guidance — general population: Ideal air quality for outdoor activities.
- Environment Canada AQHI guidance — at-risk populations: Enjoy your usual outdoor activities.

Hazard Snapshot — Overall Risk: High Risk

AIR QUALITY Low Risk 2.0 AQHI	HEAT Low Risk 18.7 °C apparent/humidex	WIND Low Risk 34.9 km/h peak gust	PRECIPITATION High Risk 2.4 mm/h maximum	THUNDERSTORM High Risk Thu 12:00 PM first forecast signal
AQHI RATE OF CHANGE Low Risk 0.0 AQHI change/hour			WIND SHEAR Low Risk 18.8 ° direction diff (10m vs 120m)	

Local Monitoring

Nearest station 2.0 Edmonton Lendrum · 3.51 km	Blended estimate 1.0 confidence High	PM2.5 (station) 4.5 µg/m³ Edmonton Lendrum · 3.51 km	PM2.5 (community) 5.6 µg/m³ ACA_LaurierHeightsCL · 1.8 km
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Prepared by Alberta Capital Airshed for Edmonton Symphony Orchestra outdoor concerts. Beta decision-support product
— for the interactive map, live cameras and hourly outlook, scan the QR code or visit [https://dkevinm.github.io/](https://dkevinm.github.io/eso_sitre/)
eso_sitre/

